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by

TIL5050 Design Project

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Preface

Preface…

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This TIL5050 Design Project has a start around november 2025

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1 Introduction

This project is carried out for NACO, an airport-focused consultancy within Haskoning, which specialises in airport design and operations (Consultancy, 2026). As Europe faces increasing pressure to reduce aviation-related emissions, a growing discussion concerns the replacement of short-haul flights with rail-based alternatives. However, this transition is complicated by conflicting interests: while shifting passengers from short-haul flights to land modes offers clear environmental benefits, it simultaneously challenges the traditional business models of airlines and airports alike, and will reduce airport revenue streams previously associated with short-distance traffic. Additionally, historical precedents have shown travellers generally highly prefer road and air transportation over rail. These tensions form a critical part of the project context.

Motivation

The project is motivated by the urgency to make European mobility more sustainable while avoiding the environmental and physical impacts of large-scale airport expansion. Short-haul flights remain a major contributor to capacity strain and emissions despite the existence of potentially viable rail corridors. The commissioner's needs statement highlights that numerous short-haul routes demonstrate large and consistent passenger flows, yet no efficient high-speed rail alternatives exist to absorb this demand. By identifying which routes are most promising for rail substitution, exploring how such rail services should be structured to capture passenger demand, and determining how airports should best proceed within the new service patterns, the project aims to support a long-term shift toward cleaner travel. This project will sit at the intersection of the aviation and transport sectors, both of which are undergoing structural transitions driven by climate policy, capacity constraints, and long-term mobility planning. The outcomes have relevance not only for reducing aviation emissions but also for improving network efficiency, strengthening multimodal transport integration, and guiding strategic infrastructure investments across Europe.

1.1. Problem Definition

The European Commission has stated in the past that it would like to reach net zero Greenhouse Gas (GhG) emissions by 2050, with an interim goal of a 55% reduction by 2030 when compared to 1990 GhG levels. To achieve this, a large decrease in the emissions caused by transport is needed, in particular for aviation, which accounts for 14% of total transport emissions (Bruno et al., 2025). The European Commission's sustainable and smart mobility strategy has set targets to double the amount of HSR traffic, hoping to provide this as an alternative to short-haul flights and cause a modal shift from plane to train. This causes a structural dilemma for the aviation sector. While climate regulations increasingly restrict emissions and airport expansions, travel demand continues to grow, causing a need for additional service. The main solution proposed is a modal shift from aviation to high-speed rail, especially for short-distance flights. However, it is unclear how this transition would be integrated in contemporary airport systems.

1.2. Literature Review

This section provides an comprehensive literature review over air-rail integration and their relevance to the design project.

1.2.1. Environmental and general impacts Air/Rail

In the first part of the Literature review, the impacts of air and rail integration are studied, as well as the additional environmental effects. The main goal of reducing flight numbers are mostly environmentally based, in today's world of growing climate concerns. The main operator of the Nordic triangle, SAS, mentions reducing their climate impact as the most important environmental priority. This is because 99% of the company's emissions comes from aircraft operations. To reduce climate impact, SAS continues to modernize their fleet with increasingly more efficient aircraft. However, they also importantly state "finding disruptive technology to be used to reduce CO2 emissions in aviation" as a priority (SAS, 2024). In their annual report on sustainability, they present the emissions from routes per distance category. In this project's scope, the flights of 500 km and less account for 11.1% of the total yearly emissions, and flights between 500 km and 800 km account for 10.2% of total emissions. This is a significant share of SAS's yearly emissions, which could be significantly reduced by replacing short-haul flights within the Nordic triangle by HSR alternatives.

The potential impacts of this HSR replacement of short-haul flights is discussed in multiple research papers. A paper from Givoni and Banister, 2006 talks about the potential replacement of flights by HSR leading to freed slots at airports, which airlines can benefit from. From the environment perspective, this could be seen as a backfire in the wrong direction, since these available slots will then be filled again, possibly by longer-distance flights that emit more greenhouse gases. In the end, this comes down to climate policies on flight numbers per airport, which could be mandated so that this backfire situation is avoided. Furthermore, Givoni and Banister, 2006 mentions that there are additional social and economic benefits to the integration of multiple transport modes, especially if this is paired with decreased emissions.

As mentioned by Janic, 2010 potential benefits could include 'savings in the quantities and related costs of social and environmental impacts such as airport airside delays, noise, and local and global emissions of greenhouse gases'. These apply best to crowded airports, that are nearing their maximum capacity. Even though Copenhagen, Oslo and Stockholm are nowhere near the numbers of major hubs like Heathrow and Schiphol (Eurostat, 2025), they still serve as major hubs for their respective countries. For this reason the aforementioned benefits can be expected in this project's scope.

Xia and Zhang, 2017 further builds on this, showing that social welfare is gained from a proper air-rail integration. However, it also mentions the lack of incentives for both the airline and railway companies. This is something that should be carefully explored during the project, since this might severely block the success of project, especially financially. The paper suggests the pre-establishment of an agreeement on sharing revenue between all parties involved, to avoid financial disbalance between the parties as a result of better air-rail integration. The further financial impact of air-rail integration can be explored by using vertical differentiation to calculate air traffic volumes, consumer surpluses of passengers and social welfare. On this, the paper says 'Consumer surplus always rises from the reduction in air-rail connecting time, reflecting, unsurprisingly, that an integrated hub with seamless transfer between air and rail services is always preferred by passengers.'

1.2.2. Key Requirements and Feasibility

Designing an air/rail integrated services comes with many key requirements. for the Environment and Infrastructure, 2020 presents the 4 layer rail system, giving more insight to the complexity of the rail transport system. The first layer consists of the mobility services, which assist travelers with planning their journey and buying tickets. The second layer consists of the transport services, e.g. the railway companies and the vehicles. The third layer concerns traffic services, including the allocation of train paths and traffic management. Finally, the physical and digital infrastructure is the fourth layer. This model gives insights in how the different layers depend on each other. Each upper layer induces demand on functions ant services provided in the layer below. Transport services for instance cannot exist without the presence of any infrastructure. Additionally, they mention four important factors listed below:

- **Passengers appreciate:** journey time, comfort, frequency, price, punctuality, safety.
- **Bookability:** being able to book long.

- **Findability:** international train services are hard to find.
- **Assurance:** passenger rights, direct connections, reliable timetables. Transfers make a journey more uncertain in various ways, such as taking control of luggage. The authors highly recommend creating 'corridor coordinators' and focusing on UE funds to be spent on specific routes.

Integrating air and rail systems requires collaboration and can lead to tension among stakeholders. These are addressed in Toet et al., 2025. They discussed 9 distinct tensions associated with air/rail integration, focusing on multimodal journeys (rather than substitution). For our project, four tensions are relevant in particular:

- **Train stop allocation:** they established that there is no air-rail integration if the train does not stop at the airport.
- **Schedule (mis)alignment:** coordinating the train and flight schedule is a key part of a multi-modal journey. However, due to different incentives, airlines serve international travelers, whereas railways act primarily on the national level.
- **Booking systems:** To book a seamless journey, connectivity between the booking systems of airlines and railways is required.
- **Different passenger experiences between air and rail travel:** At an operational level, this included the travel time accepted by passengers. Passengers taking an air-rail integrated trip may view a long rail leg as an inferior alternative to a flight, rather than an integrated part of the journey. They typically accept less travel time than a rail passenger.

In Thailand, there is ongoing integration between air and rail, referred to as the Bangkok Airport Rail Link (ARL). Rattanakijsuntorn et al., 2024 conducted an assessment on this development. They discuss the following advantages:

- **Travel time can be minimized**, mainly for the connection between the airport and the city center.
- **The integration of tickets**, baggage handling and aligned schedules lead to a high level of service.
- **Modal shift from air to rail leads to less emissions** (only if the freed up airport slots are not replaced by other flights).
- **Settlement** near train stations.

The authors point out that there are still three crucial operational considerations that must be taken into account to complete air–rail integration, despite the fact that all of the railway tracks and operating trains have been provided in the article. These include passenger information, baggage check-in facilities, and air–rail ticketing. This also raises potential issues concerning working with different levels of governance, in the case of this project three different countries.

Some articles also discuss the feasibility and added value of air-rail integrated projects. Royal HaskoningDHV, 2025 investigated the international train connection via the Lelylijn and Wunderline from the Netherlands to North-Germany and Scandinavia. They put this project in a broader perspective and also investigated alternative corridors. Al of these lead to more added value in terms of travel time savings and yearly n.o. international passengers, as well as lower investment costs. Scenarios with (partially) new corridors result in more added value compared to upgrading existing infrastructure. However, this is only the case when investing in one single corridor, rather than 'double' investments.

1.2.3. Passenger Preferences

For this design project relevant passenger preference parameters are important to retrieve. Based on provided literature, these can be obtained.

The master thesis van Dam, 2024 provides us with a stated-choice experiment. This study investigated the research question using a stated-preference survey among 541 Dutch respondents from the Dutch Railways (NS) panel. Participants evaluated various air–rail combinations through rating experiments that focused on both service and comfort attributes. They were asked to assess the attractiveness of different configurations. These attributes (service and comfort) were subsequently incorporated into a mode choice experiment alongside operational factors such as travel time and travel cost. The collected data were analyzed using

several discrete choice models, including the multinomial logit model and the latent class choice model, to estimate parameters and identify distinct consumer segments within the air–rail market (van Dam, 2024).

A central finding from the service rating experiment is that integrated ticketing is by far the most valued feature (0.554). It is rated more than twice as attractive as integrated information provision (0.213) and roughly three times as attractive as integrated luggage handling (0.185) and integrated customer service (0.177). In practical terms, offering integrated luggage handling, a unified information platform, and integrated customer service together yields approximately the same perceived attractiveness as offering integrated ticketing alone. When no service integration is provided, respondents assign an average attractiveness score of 2.4 out of 5. By contrast, full integration of all evaluated services results in an average score of 4.7 out of 5. Thus, maximum service integration is perceived as nearly twice as attractive as having no integration at all (van Dam, 2024). This thesis report also contains theoretical maximum substitution rates for HSR replacement of flgihts, with a maximum air-rail share of 70 procent can be achieved. This shows that there is a willingnes for passenger to use HSR instead of short-haul flights and HSR integration with flights.

A European Union report (Commission et al., 2025) also provides a study in preferences for air-rail services. A significant portion of the report focuses on multimodal travel, specifically the integration of collective transport modes like air and rail within the European Union. At the current moment, combining an airplane and a train is the second most common multimodal travel combination in the EU, used by 36% of respondents who reported combining transport modes. In Sweden, 63% of the respondents used the train for long-haul regional trips in the past 2 years, compared to 43% using a plane. In Denmark, the percentages are 47% and 35% respectively (Commission et al., 2025). The report also states that for those who have experience combining different collective transport modes (such as a flight with a rail service), over a third find it difficult to book (30% "rather difficult" and 5% "very difficult"). Approximately 31% of total respondents have never booked a journey that combines different transport modes, suggesting a significant potential for growth if digital integration improves (Commission et al., 2025).

More quantitative passenger parameters can be found in literature. For example, an important parameter in the design process of an air-integrated rail service is the Willingness to Pay (WtP). Chiambaretto et al., 2013 found various WtP values in a study of airports and HSR in France and utility contributions of various attributes (transfer time, service quality, ride time, etc.). They found high utility contributions for low connecting transfer times. However, they also concluded a significant difference between leisure passengers and business passengers in parameter values (Chiambaretto et al., 2013). A study by Bruno et al., 2025 found that travel times losses are limited up until 5:30 hours of train journey time.

A 2024 study conducted by Román and Martín, 2014, looked for WtP values for air-rail integration in Spain by constructing a stated choice experiment. They found range of WtP values for attributes like access mode, egress mode, trip purpose, age, in-vehicle times and transfer times. Their results showed that passengers value in a similar way the complete fare integration or the soft-integration whereby an insurance company could reduce the uncertainty of the monetary losses for potential delays (Román & Martín, 2014). The authors Xia and Zhang, 2017, showed that long connection times are not desired by passengers, especially for business travelers, also stated by Buire et al., 2023.

1.2.4. Schedule Design

From the previous section 1.2.3, it was shown that short transfer times and good air-rail synchronization are important attribute preferences for passengers. Therefore, for this design it is important to model a robust air-rail synchronization design. The authors Buire et al., 2023 provide an air-rail timetable synchronization model for improving passenger connections in Europe within and across transportation modes like high-speed rail and conventional rail lines. Their model is a Mixed Integer Linear Programming (MILP) model that aims to minimize the total multimodal passengers' transferring discomfort, while, as a secondary objective, keeping schedule deviation low (Buire et al., 2023). With their model, they found results indicating that adjusting just 13% of European flights by an average of 11 minutes can boost the number of suitable passenger connections by 60%. Furthermore, when both airlines and railway operators coordinate their schedules, passenger comfort improves and operator costs decrease; yielding greater benefits than when changes are made unilaterally (Buire et al., 2023).

A paper by Ke et al., 2020, developed a MILP formulation for an integrated rail-air synchronization timetable. Although multiple optimization models can be found for rail timetables (Kang et al., 2018; Wen et al., 2024),

the literature for rail timetable integrated with air operations optimization is limited (Ke et al., 2020).

In their paper, the authors try to maximize the synchronizations of rail and airplanes (by defining a synchronization by a specified time-window). Their (multi-objective) model includes parameters such as travel times, preferred time windows, passenger preferences for business and leisure passengers. Their model is especially useful for designing a robust timetable for rail to link efficiently with airports. The authors applied their model to an east China case study and found a 24% increase in synchronizations compared to the default rail timetable (Ke et al., 2020).

1.2.5. Design Method

Why does method Choice Matters in Complex TIL Design?

1.3. Research and design questions

What has not yet been researched are the main reasons why in general short-haul flights exist. This information is essential to create a viable rail alternative for short-haul flights. Additionally, the practical side of integrating air and rail is still open to study. It is still largely unknown what such a journey would look like.

Extending state-of-the-art knowledge about air-rail integration to the functionality of an airport, such as turning it into a multimodal hub, could be a promising direction. For this project, air-rail integration would involve the substitution of short-haul flights by rail. By integrating the two modes of transport, the rail journey would serve as a viable connection to passengers arriving with long distance flights that would only fly between main hubs.

To address the formulated problem and the research gap, this project poses the following main question, as well as some sub-questions:

- What should the design of an integrated air-rail service look like to effectively replace short-haul flights within a hub-based airport system?
    1. What functional and economic roles do short-haul flights currently fulfill?
    2. What are the interests, incentives and constraints of key stakeholders in an air-rail integration project?
    3. What performance and functional requirements should rail meet to provide an effective alternative to short-haul flights, considering the respective stakeholders?
    4. How should the air-rail service be designed in terms of infrastructure, airport operations and passenger journeys?
    5. What are the expected impacts on airport and rail capacity allocation, emissions and airport revenues?

1.4. Approach and Scope

To answer the research and design questions, the project starts with an investigation on short-haul flights. This entails the main reasons for their existence, as well as the routes that are served by this type of flights. Additionally, our goal is to find the functional requirements associated with flights and potential substituting rail services. Once these factors are known, we can continue with the design part of the project. It starts with modeling the flight and rail network around an airport that is used as a case study for the design. Based on preliminary deliberations and discussions with our supervisors, we found three corridors that could be interesting to look at, which are the following.

- The Amsterdam - Cologne - Frankfurt corridor.
- The Nordic triangle: Copenhagen, Stockholm, Oslo.
- The hub of Warsaw.

After conducting further interviews with Barth Donners and Remco Troquete, both Consultants of Sustainable Mobility at NACO, we were provided with further advice on choosing a suitable case study and a reason-

able scope. It was thus decided to choose the Nordic Triangle as the geographical area of our design; thus, we only focus on short-haul flights from the respective airports within that area. The Nordic Triangle was chosen because the current level of rail service between Copenhagen, Stockholm, and Oslo is generally considered to be in need of improvement and the fact that SAS is the main joint flag carrier among the three airports, which should somewhat simplify the stakeholder analysis with respect to the airlines.

For reproduction purposes, we use publicly available flight and demand data. We will select 2019 to avoid distortions caused by the pandemic and its after-effects. These data are separated for every month, but we decide to aggregate the demand and flights over a full year, because aggregation prevents seasonal bias and allows for a structural analysis, rather than peak driven. Lastly, the data do not differentiate between types of travelers, leading to the assumption of one generic passenger type for the modeling part. This simplification limits heterogeneity, but allows for a network-level focus. Since the research part about short-haul flights can give insight into travel motives and target groups, we still incorporate these into our final design choices.

2

Current situation

2.1. Context Analysis

This chapter provides a context analysis of the current transport landscape within the Nordic Triangle. It begins by evaluating the preexisting rail infrastructure of the three primary hubs: Oslo Gardermoen (OSL), Copenhagen (CPH), and Stockholm Arlanda (ARN). These sections detail the geographical and commercial positioning of each airport, specifically their roles as multimodal nodes that already facilitate high-speed ground transport through dedicated lines like the Gardermoen and Arlanda corridors.

Afterwards, the chapter goes into a stakeholder analysis, identifying the primary interests, challenges, and power dynamics of the actors involved. By framing these stakeholders within the constraints of a market-based economy, the context analysis establishes the baseline from which the design project seeks to identify viable opportunities for modal shift.

2.1.1. High Speed vs. Normal Speed Rail

According to the European Union directive 2016/797 (regarding the interoperability of the trans-European high-speed rail system), HSR lines are classified into three distinct categories to standardize technical specifications: Parliament and Council, 2016

- **Category I:** Specially built high-speed lines equipped for speeds generally equal to or greater than 250 km/h.
- **Category II:** Specially upgraded high-speed lines equipped for speeds of the order of 200 km/h.
- **Category III:** Specially upgraded high-speed lines which have special features as a result of topographical, relief, or town-planning constraints, on which the speed must be adapted to each case.

While some may consider only Category I as "true" high-speed rail, the overarching objective of HSR is to achieve short, competitive travel times between important destinations. To that extent, those objectives can just as well be achieved by lower-speed rail as well. Tove Andersson, 2013 To achieve the necessary average speeds for market competitiveness, HSR infrastructure requires significant technical precision; tracks must be smoother than conventional lines, featuring larger curves, refined surfaces, and minimal gradients. To allow for these high speeds, the train tracks have to be significantly smoother than regular train tracks; larger curves, smoother surface and smaller slopes. This characteristic makes the price of both construction and maintenance significantly higher than for regular rail. According to Branković and Kalem, 2021, building costs of high-speed rail infrastructure largely varies per country and project. They mention some example projects in southern Europe, where costs per kilometer vary from 9 million euros to 68 million euros. They mention that terrain and land acquisition largely influence the project costs. This is because a mountainous terrain requires the construction of many bridges and tunnels, which are more expensive than building along an existing highway on flat land, especially for high-speed infrastructure.

2.1.2. Railways in the Nordic Triangle

Operationally-speaking, the historical development of rail transportation within the Nordic Triangle has evolved from a series of state-monopolized national networks toward a more liberalized, market-driven landscape. Throughout the 19th and 20th century, the railway systems of Norway, Sweden, and Denmark matured largely in isolation, prioritizing domestic industrial transit and national connectivity over seamless cross-border integration. However, starting in the late 1980s and 1990s, the Nordic countries became early adopters of vertical separation, unbundling infrastructure management from train operations to foster competition. This separation between operations and infrastructure will be noted when considering the stakeholder analysis.

Infrastructurally-speaking, the late 1980s and 1990s were marked by flagship projects such as the Øresund Bridge, which physically linked the Danish and Swedish networks, and the 64 km Gardermoen Line, Norway's premier (and only) high-speed corridor. These investments served to greatly increase the connectivity in the region.

Despite these investments in infrastructure, a primary obstacle still remaining is the divergence in signaling systems; while much of the network historically utilized Automatic Train Control (ATC), Denmark's transition to the European Train Control System (ETCS) requires specific rolling stock compatibility to maintain cross-border operations.

Furthermore, the electrification of the Triangle remains fragmented, as Denmark operates on a different voltage than Sweden and Norway, necessitating the use of dual-voltage trains to bridge the gap. These technical complexities, combined with the varying track requirements of the Scandinavian terrain, result in current rail journey times between all three cities of six to eight hours. Consequently, rail continues to struggle against the deep-seated market dominance of short-haul aviation, which currently offers flight durations of around one hour between the three capitals.

2.1.3. Current Travel Status in the Nordic Triangle

In the sections below, the three main airports in the Nordic Triangle will be presented. These are the three capitals defining the Triangle, being Oslo, Copenhagen and Stockholm. All three airports are serviced by train stations integrated into the airport's design in some way, connecting them both to the city it serves, as well as the surrounding areas. All three airports are also connected with each other by frequent flights, mostly operated by SAS Eurostat, 2025. Another airport in this Nordic triangle is Gothenburg Landvetter Airport, which lies on the Oslo-Copenhagen axis. This is a significant node in the area, and will be taken into account in the design, but an analysis on this airport is deemed beyond the scope of this project and therefore will not be provided.

Modes of transport vary greatly among the areas in the triangle, both per country, but also per area of the country. Naturally, micromobility and public transport thrive in the bigger cities, whereas car use is more predominant in rural areas. With all airports being well-connected to the surrounding rail network, local trips to and from the airport are already frequent. However, train trips between the 3 airports are severely lacking. With no direct services taking at least 6 to 8 hours, depending on the direction, the train alternative is not able to compete with the current flights connecting the airports. An improvement in both the service and the infrastructure related to the trains between these airports, could help with the successful replacement of the existing flights.

2.1.4. Oslo Gardermoen Airport (OSL)

Oslo Gardermoen Airport (hereafter referred to as OSL by IATA code (IATA, 2026))[1] is the primary international airport in Oslo and the second largest in Scandinavia and the Nordics, processing approximately 27,000,000 passengers in 2025, of which 12,500,000 were coming to/from international destinations. (AS, 2025)

From its initial design phase, OSL was planned to have direct rail access to the Oslo city center. It was modeled as a multimodal hub, utilizing the 64 km Gardermoen Line, Norway's only high-speed railway, which allows for speeds up to 210 km/h. This infrastructure enables one of the world's highest public transport shares for an airport, with nearly 70% of passengers arriving via rail or bus. AS, 2016

[1] Later, the ICAO codes for the airports is also used (ICAO, 2026)

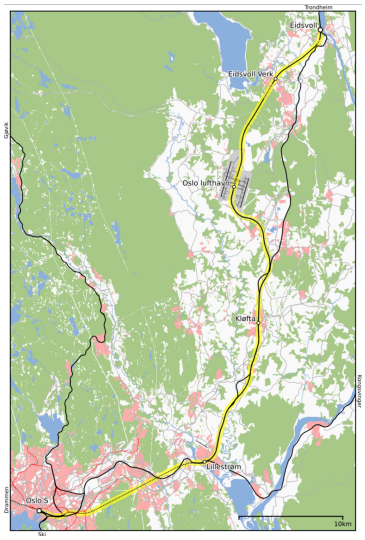


Figure 2.1: Map of the Gardermoen Line (highlighted) and Oslo Gardermoen Airport (Modified from OpenRailwayMap, 2026)

The airport's train station itself plays an integral role in the overall network of eastern Norway, with multiple intercity and local routes making use of it. As such, the infrastructure for a competitive rail alternative is already partially established; Swedish state railways (SJ) already operate intercity services directly to the airport. The existing use of Type 71 and Type 73 tilting train sets on the Gardermoen Line demonstrates that the technical and operational precedent for high-speed rail exists, even if current long-distance rail journey times are struggling to compete with the convenience of short-haul flights.

Railway infrastructure notwithstanding, the current market preference inside the Nordic Triangle remains dominated by aviation. The strategic importance of OSL is mainly due to its position within the Nordic Triangle, where it maintains high-frequency links to Stockholm (ARN) and Copenhagen (CPH). These two air routes are dominant within the range of international destinations served from OSL, and are among the busiest in the European Economic Area, each consistently serving over one million passengers annually (eurostat-2025).

Beyond transit, OSL is a critical commercial engine for the airport's state-owned operator, Avinor. The airport has an extensive retail footprint, which generates roughly half of the operator's income, which is then used to cross-subsidize smaller regional airports across Norway. This economic reality means any proposed shift to rail must account for the airport's role as a financial stabilizer within the Norwegian market. Furthermore, the terminal's has a couple points of contention for passengers passing through the airport. The manual customs clearance process for international-to-domestic transfers remains a significant deterrent for transit travelers.

2.1.5. Copenhagen Airport (CPH)

Copenhagen Airport (hereafter referred to as CPH by IATA code (IATA, 2026)) is the primary international airport for Copenhagen. The catchment area also captures the wider Øresund region, which includes Zealand, Malmö and the province of Scania, in the south of Sweden. CPH is the largest air hub in Scandinavia and the Nordics, with 32,400,000 passengers being handled in 2025 How many pax to international and domestic destinations.

Since its inception in 1925 (making it one of the worlds oldest civil airports), the airport has benefited from a high degree of intermodality through direct rail and bus. Its rail station, located directly beneath Terminal 3, connects CPH with Copenhagen Central Station in approximately 12–15 minutes, with trains operating at a frequency of every 10 minutes during the day. There are of course further connections, which extend into Sweden, notably to cities like Malmö and Gothenburg, reinforcing the airport's pivotal role as a multimodal regional node, by giving direct access to the rail network of Sweden and Denmark. Furthermore, CPH gives travellers access to the metro network, which offers access to the cities transport network. Combined, these systems underpin a highly competitive ground transport offering that mirrors the Nordic model of airport integration with wider regional mobility.

The strategic importance of CPH within the Nordic Triangle is reinforced by its extensive route network. CPH offered 366 routes to 171 destinations in 2026, driven by the expansion of SAS, Norwegian, and Ryanair which together, carry 62% of all passengers. CPH also saw the introduction of new long-haul services to destinations such as Minneapolis, Seattle, Nuuk, and Seoul, strengthening intercontinental connectivity and its position as the principal transfer point for Denmark and southern Sweden. Transfer traffic rose sharply to 3.3 million passengers, a 23% increase compared to the year before, demonstrating the airport's role as a major redistribution hub for northern Europe (Orban, 2025).

CPH is a significant economic engine for the operating organisation, Copenhagen Airports A/S. According to own performance figures, in 2024 the airport generated DKK 5,070 million in total revenue, with roughly 40% of that revenue being derived by other activities, not directly relating to air-operations, rather commercial offerings such as retail and food, as well as extra services like parking and hotels. Although retail capacity has been temporarily constrained by the ongoing Terminal 3 expansion (scheduled for completion in 2028), concessions remained a substantial contributor to profitability. The airport recorded DKK 1,339 million in profit before tax, reflecting robust demand and strong earnings from both aeronautical and commercial revenue streams Copenhagen Airports A/S, 2025.

Alongside its operational and commercial expansion, CPH is pursuing an ambitious sustainability program, targeting Net Zero airport operations by 2030. Initiatives include transitioning ground vehicles to HVO biodiesel and sourcing 100% renewable electricity from offshore wind. The airport notes that aviation emissions remain the sector's largest challenge, emphasizing the need for broader policy support to scale sustainable aviation fuels.

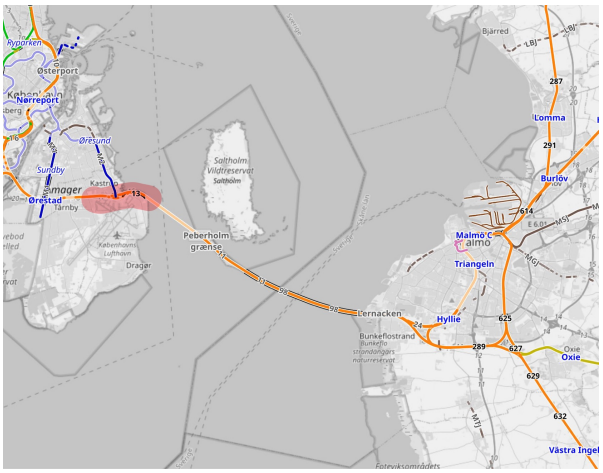


Figure 2.2: Map of the Copenhagen Airport and its railway connection (highlighted in red) and the surrounding railways (Modified from OpenRailwayMap, 2026)

2.1.6. Stockholm Arlanda Airport (ARN)

Stockholm Arlanda Airport (IATA Code: ARN) is the main international airport serving Stockholm and the largest airport in Sweden. As of 2025, it serves roughly 32,000,000 passengers per year, of which 25,000,000 had international destinations, and 7,500,000 were domestic. Airports, 2026

Stockholm is served by one other commercial airport in the municipal area, Bromma Airport. However, due to aftereffects from the COVID-19 pandemic, Bromma will soon be closing, with all current flights being rerouted to serve Arlanda instead. This will leave Arlanda as the only major airport serving the cast majority of passenger traffic in the Stockholm region, further increasing its importance. Markus Hjortsman, 2024

In terms of railway connections, Arlanda Airport is served by the Arlanda Line, a 19 km branch of the Swedish East Coast Line that includes three dedicated underground stations: Arlanda South, Arlanda North, and Arlanda Central (Arlanda C). The airport express service, Arlanda Express, operates at speeds up to 200 km/h, providing a 18-minute link to Stockholm Central. Crucially, Arlanda C serves as a major node for the national rail system, where long-distance SJ Intercity and SJ High-Speed trains (X2000 rolling stock) connect the airport directly to Uppsala, Gävle, and southward toward the other vertices of the Nordic Triangle.

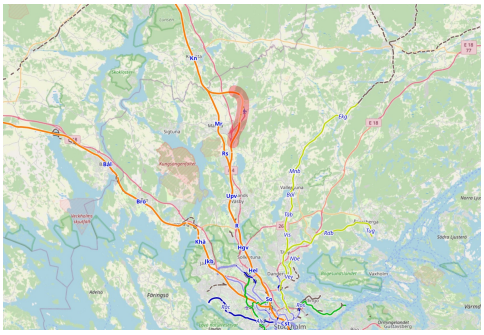


Figure 2.3: Map of the railways north of Stockholm, with the section including all three Arlanda stations highlighted in red (Modified from OpenRailwayMap, 2026)

Commercially, ARN operates under the state-owned Swedavia group, which utilizes the airport's high-margin retail and real estate revenue to cross-subsidize less profitable regional airports throughout Sweden. At the heart of this commercial strategy is SkyCity, a massive landside complex situated between Terminals 4 and 5. By housing hotels, offices, and retail, SkyCity capitalizes on the massive foot traffic generated by the rail stations. This market-driven interdependence means that any shift in passenger volumes from air to rail must be balanced against the economic stability of the national airport network, as Arlanda is the primary financial engine for Swedish aviation infrastructure.

2.2. Stakeholder Analysis

- **Airports**
  - **Interest:** Airports act as multimodal hubs and have a vested interest in maintaining their competitive advantage over other modes of travel. While shifting short-haul passengers to rail challenges traditional revenue streams from traditional revenue traffic, it will simultaneously free up constrained slot capacity at some overburdened airports.
  - **Challenges:** Developing a compelling business case to remain commercially viable as a multimodal hub. Additionally, given current growth rates in the aviation industry, without compelling alternatives, they must cope with growing demand and capacity limits without large-scale physical expansion, which is often restricted by space and noise regulations.
  - **Power:** This actor has high power as the provider of infrastructure and the manager of slot reallocations, which could be shifted from short-haul to long-haul flights.
- **Airlines**
  - **Interest:** Airlines focus on operational efficiency and meeting passenger demand. They are interested in how air-rail integration can support their business models, particularly for connecting long-haul markets. Some airlines already have integrated ticket booking systems with railway operators (e.g., Deutsche Bahn and Lufthansa).
  - **Challenges:** Shifting to rail will challenge traditional business models for short-haul routes. Airlines require rail alternatives that are competitive in terms of travel time, connectivity, and convenience to satisfy their passengers. With respect to ticket integration, there are bureaucratic obstacles to companies integrating their systems.
  - **Power:** Airlines hold high power as they own the aircraft fleets and decide which flight corridors to maintain or cancel based on commercial viability.
- **Railway Operators**
  - **Interest:** These agencies (e.g., NS, Eurostar, DB) are very interested in capturing the modal shift from plane to train. Their goal is to maximize passenger volumes and provide services that meet the requirements of former aviation passengers.

  - **Challenges:** They must provide a level of service (frequency, convenience, late-night departures) that can effectively compete with airlines.
  - **Power:** They have high power in the decision-making process as the actors performing the actual ground-based transport operations.

- **Railway Infrastructure Operators**
  - **Interest:** These entities (e.g., DB Infra, ProRail) manage the track capacity and technical feasibility of the rail network. Their interest lies in the long-term planning and optimization of rail corridors, and the expansion of said corridors.
  - **Challenges:** Existing corridors may either lack the capacity for increased HSR traffic, or require a major overhaul, or must be simply built as a greenfield development. All three options would require significant investment to become viable alternatives to flights.
  - **Power:** They hold a high amount of power, as they are the maintainers and planners of the physical infrastructure required for the modal shift.

- **European Ministries**
  - **Interest:** National ministries represent the interests of individual member states in infrastructure projects. They are interested in domestic benefits, economic impact, and meeting EU environmental targets.
  - **Challenges:** They face trade-offs between international gains (network efficiency) and domestic benefits or costs associated with building new rail lines.
  - **Power:** Medium power, while they decide how projects are funded, their influence is limited by market demand, as they cannot force a modal shift if the rail alternative is not preferred by users

- **European Commission**
  - **Interest:** The Commission aims to reach net-zero Greenhouse Gas emissions by 2050. Their strategy includes doubling HSR traffic to reduce the 14% of transport emissions caused by aviation.
  - **Challenges:** They must balance European-wide directives and environmental regulations with the economic interests of individual member states and transport sectors. Historically, the potential for effective policy-making to drive modal shifts has been low across most corridors.
  - **Power:** Medium power; they provide the regulatory framework and goals, but they cannot override the market mechanism or individual consumer choice

- **Passengers**
  - **Interest:** Being the primary deciders and payers, passengers are the main actor class. They mainly consist of local market travellers, whose demand can easily be substituted, and long-haul travellers who use short-haul flights as connecting legs. Their primary interests are primarily split between travel time, price, connectivity, and convenience, with many currently preferring air and road transport.
  - **Challenges:** For rail to be a viable alternative, it must have convenient departure times and high levels of service. Without significant improvements, passengers will continue to exercise their market preference for existing air or road options. Business travellers especially deem late-night departures and frequent departures to be important. Passengers also expect a seamless experience during air-rail integration.
  - **Power:** High power. As the ultimate decision-makers who pay for the service, their willingness to shift is the single most critical factor. They hold the power to determine the success or failure of the entire modal shift project.

- **Commercial Operators**
  - **Interest:** This group consists of the companies who will be operating the commercial properties inside the airports, located both landside and airside. They encompass retailers (grocery stores,

clothing shops and duty free shops, etc), service providers (banks, pharmacies, etc) and restaurants, among others. They pay the airports for the leasing of space and also a percentage of their sales.

  - **Challenges:** Their main concerns will be to balance their money spent on leasing space and making their money back from customers. They would prefer to have as much foot traffic as possible, with
  - **Power:** Airports derive a relatively large percentage of their revenue from money spent inside retail establishments on their property, thus the commercial operators have a large amount of power over the overall design of the airports and the amount/type of services offered as well.

- **Non-Users**
  - **Interest:** This group comprises the residents living in areas near the airports and proposed rail corridors affected by the infrastructural changes. Their primary interests are the reduction of air and noise pollution and the overall economic impact on their local community.
  - **Challenges:** While shifting to rail aims to reduce aviation-related emissions and noise, the construction of new corridors can lead to local environment and physical impacts during the transition. They may perceive a trade-off between European-wide gains and immediate domestic disruption.
  - **Power:** Non-users generally have low direct power over technical design but can exert influence through local ministries, environmental NGOs, and public consultation processes.

2.3. Relation Between Stakeholders

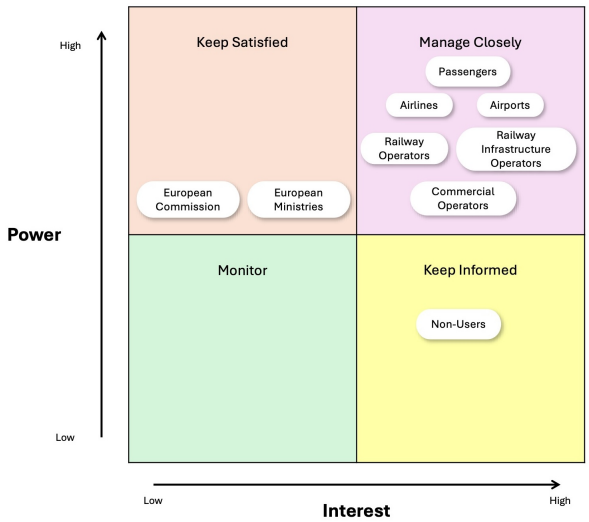


Figure 2.4: Power Interest Matrix

Figure 2.4 shows the power interest matrix with the relationships between the stakeholders for this design project.

3

Design Method

3.1. Seven-stage model of Soft Systems Methodology

As stated in section 1.2.5 the design methodology used for this design project is the Seven-stage model of Soft Systems (Checkland, 1999). A schematic is given in figure 3.1. This methodology gives a structured 7-stage way for a design process. The seven stages are:

1. Problem situation considered problematic
2. Problem situation expressed
3. Root definitions of relevant purposeful activity systems
4. Conceptual model of the systems named in the root definitions
5. Comparison of models and real world
6. Changes systematically desirable, culturally feasible
7. Actions to improve the problem situation

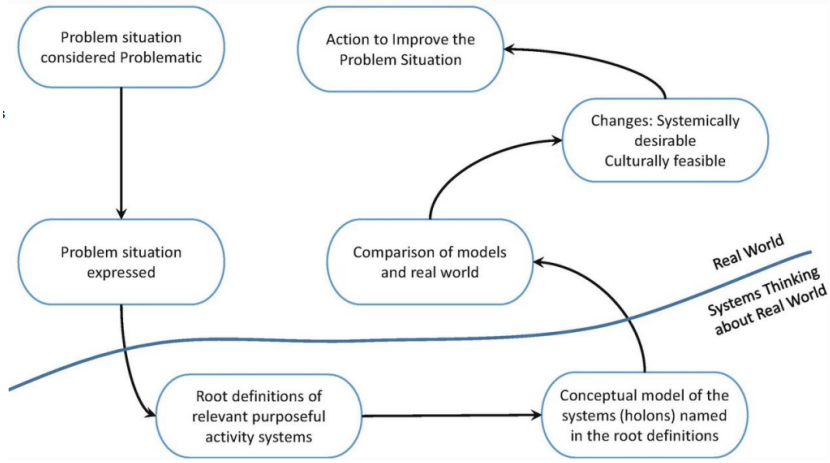


Figure 3.1: The Seven-stage model of Soft Systems Methodology (Checkland, 1999)

3.1.1. Problem Situation

The first stage of the Seven-stage model of Soft Systems Methodology is an overview of the problem situation considered problematic. In the case of this design project that would be the problem stated in section 2.1.3. Section 2.1.3 states that in the context of this project, the problematic situation concerns the current imbalance between short-haul aviation and rail transport within the Nordic Triangle, defined by the cities of Oslo,

Copenhagen, and Stockholm. Despite the relatively short distances between these metropolitan regions, air transport currently dominates passenger flows between them. As discussed in Section Current Travel Status in the Nordic Triangle, frequent flights connect the airports of Oslo Gardermoen (OSL), Copenhagen (CPH), and Stockholm Arlanda (ARN). Flight durations between the cities are typically around one hour, and airlines operate multiple departures per day. In contrast, existing rail services between these cities require approximately six to eight hours of travel time and often involve transfers, making them far less competitive with air transport.

3.1.2. Problem situation expressed

The second design stage is the current problem situation expressed. This is best done via the stakeholder analysis given in section 2.2. Rather than focusing purely on technical limitations, this stage emphasizes the socio-technical nature of the problem, highlighting how different stakeholders influence the current transport system in the Nordic Triangle and how their interests interact. Within the context of integrating high-speed rail with short-haul aviation in southern Scandinavia, several key stakeholder groups can be identified. These include airports, airlines, railway operators, railway infrastructure managers, governmental institutions, passengers, commercial operators within airports, and non-users affected by infrastructure development. Each of these actors has distinct interests, challenges, and levels of influence that shape the current transport landscape. A more detailed overview of the stakeholders is given in section 2.2.

3.1.3. Root definitions

For the root definitions it is important to have a clear view of the requirements for the design. The specific requirements are taken from the literature study as stated in section 1.2.2. Section 3.2 gives a concise overview of the relevant requirements.

3.1.4. Conceptual Model

In this stage of the design process a first conceptual model of the design is made. This model is designed with all the requirements as design constraints. In the context of Soft Systems Methodology, a conceptual model does not represent a detailed technical design or a finalized solution. Instead, it represents a structured set of activities that a system would need to perform in order to achieve the transformation described in the root definitions. The conceptual model therefore functions as a logical representation of how the system should operate in order to address the problem situation (Checkland, 1999).

However, in the context of this design project this stage will be a bit modified. In the normal design method, the conceptual model does not represent a technical design. In the case of this design project, the coneptual model will be a bit more technical detailed. That is due to the fact that with a detailed conceptual model the design can be better compared with the KPI's, as explained in the following section (section 3.1.5).

By designing the conceptual model the first step is too analyse the current situation and see where the current situation does not furfill the requirements. By addressing these points (like upgrading small rail infrastructure for reducing journey time) a conceptual model can be made which later can be compared to the current situation.

3.1.5. Comparison of Models

The forth stage of the design process is a comparison of the conceptual model versus the current situation. The comparison of the conceptual model with the current situation will be done via the KPI's as described in section 3.4. The purpose of this stage is not to determine whether the conceptual model is "correct," but rather to identify differences between the current transport system and the activities and structures proposed in the conceptual model. Through this comparison, potential improvements and areas of change can be identified that may help address the original problem situation. In this design project, the comparison between the conceptual model and the current transport system within the Nordic Triangle will primarily be conducted using a set of Key Performance Indicators (KPIs). These KPIs, which are described in Section 3.4, provide measurable criteria that allow both the existing system and the proposed conceptual model to be evaluated in a consistent and objective manner. By quantifying system performance, the KPIs make it possible to assess how well the conceptual model addresses the shortcomings identified in the current transport situation.

3.1.6. Changes to the model

The sixth stage of the design process focuses on identifying changes that are both systematically desirable and culutiral feasible. In this stage, the results from the comparison between the conceptual model and the current real-world situation are interpreted to determine which improvements could realistically be implemented (like upgrading existing rail infrastructure, synchronizing time tables with air flights, etc.). The objective is not to redesign the entire system from scratch, but rather to identify a set of changes that meaningfully improve the current situation while remaining acceptable to the involved stakeholders and feasible within existing institutional, economic, and infrastructural constraints and requieremnts

Within the context of this design project, the comparison performed in section 3.1.5 highlights several points where the current transport system within the Nordic Triangle does not meet the desired performance levels. In particular travel times between the major cities remain significantly longer by rail than by air, transfer processes between transport modes are often not seamless and service freuqencies on long-distance rail corridors remain insufficient to attract passengers who currently rely on short-haul flights. These shortcomings indicate that targeted improvements in both infrastructure and service design are necessary for rail to become a competitive alternative.

Based on these observations the conceptual model can be refined by introducing changes that improve the systems performance while considering the interests of the stakeholders identified in the stakeholder analysis. Possible changes include the upgrading of existing rail corridors to support higher operating speeds, the construction of limited sections of new high-speed infrastructure on critical bottlenecks, and the improvement of timetable coordination between long-distance rail services and airport operations. Additionally, improving intermodal integration at airports such as through coordinateded ticketing systems optimized transfer time and integrated passenger information systems can enhance the overall passenger experience and make combined air–rail journeys more attractive.

3.1.7. Actions to improve the current situation

The last stage in the seven stage model of soft systems methodology is identifying the actions to improve on the current system. In the context of this design project, the goal of the proposed actions is to improve the integration between rail and air transport within the Nordic Triangle and to reduce the dominance of short-haul aviation on routes where rail could serve as a competitive alternative. Based on the comparison of the conceptual model with the current situation and the identification of feasible improvements, several type of actions, changed and improvements can be identified and be concluded.

3.2. Requirements Specification

In this section, the system requirements are specified. Based on the literature investigation in Subsection 1.2.2, six key requirements for the design are proposed.

Journey Time

Journey time is one of the key aspects to minimize in an air-rail integrated journey (Toet et al., 2025 & for the Environment and Infrastructure, 2020). Since the train leg is only part of the air-rail integrated journey, passengers typically accept less travel time, compared to a standalone train trip. According to Bruno et al., 2025, air to rail substitution is justifiable up until 5:30 hours of train travel time, as this just prevents losses in generalized travel times. This is the first requirement for the design; the total journey time by train between the airports of Copenhagen, Oslo and Stockholm may not exceed 5:30 hours.

Transfers

Additional transfers are negatively valued (Chiambaretto et al., 2013). If one were to take a short-haul flight, this would also be a direct connection. Therefore, we require the train connections to be direct between airports of Copenhagen, Oslo and Stockholm. Direct connections to other main stations are included as KPI's, further elaborated on in Subsection 3.4.

Regarding transfer time, a lower bound needs to be set to account for boarding a flight, including check-in, customs and security. We use the minimum recommended arrival time for Copenhagen airport as minimum transfer time, which is two hours. Beyond that, a lower transfer time is associated with higher added value (Román & Martín, 2014).

Schedule synchronization

Schedule synchronization is partially related to transfer time, but also involves frequency and capacity. To establish a 100% substitution for short-haul flights, the design should be able to accommodate at least the same capacity. In other words, for each seat on each substituted flight, a train seat should be available. Additionally, in order to provide a viable alternative to short-haul flights, the frequency of the trains connecting the three airports should not be lower. This ensures that transfers/connections are not lost due to the substitution. An upper bound for this functional requirement regarding the schedule is the track-capacity on the railway line, which cannot be exceeded.

Ticket integration

Integrated ticketing is another key aspect of booking a seamless journey and increase level of service (Toet et al., 2025, Rattanakijsuntorn et al., 2024). To fulfill this, the passenger should be able to buy one integrated ticket for multiple journey legs. Additionally, to make booking a more user-friendly process, tickets for the journey should be applicable for all trains on the selected trajectory and day. This is to facilitate transfers, especially in case of delays in both the planes and the trains.

Airport - city connections

One of the goals of this project is to better connect the main airports of the Nordic triangle. With this, the goal of reducing short-haul flights can be achieved. However, to make the train service more attractive, the services should also pass through the central stations of the corresponding cities. This way, not only people transferring between the airports can easily use the service, but also people landing in one airport wanting to go to another city. Next to this, the train could also stop in intermediate stations to service the corresponding cities. The train can then pick up more demand along the way, which could improve its overall efficiency and financial viability. However, there does have to be a balance in this. Bruno et al., 2025 mention that there is an upper limit of 5.5 hours for train connections replacing short-haul flights, as mentioned before. The lower limit that then will be set is that at least the three airports and central stations of Copenhagen, Stockholm and Oslo have to be connected, with the possibility to connect more cities in between.

Rolling stock compatibility

As another requirement, the rolling stock should be compatible between the three countries involved. This means that the track gauge should be consistent, which is the case ("OpenRailwayMap", n.d.). As for signaling, almost all of the network uses ATC, with the exception of the Danish part, which will soon switch to ETCS. This means that the rolling stock will have to be compatible with both of these signaling systems. This is possible, since there are already trains running between Denmark and Sweden, so this is a matter of using a similar system. The same issue rises with electrification, where again Denmark uses a different voltage. This is also already fixed by the current trains, which use two different pantographs to bridge this issue. Therefore, for these requirements to be met, the current trains running on the bridge between Copenhagen and Malmö should be used as a leading example in terms of technology.

3.3. User Story

User perspective

As a passenger using the rail service within the Nordic Triangle, a local traveller, or a transfering passenger between long-haul flights, I desire a direct rail service between Copenhagen Airport, Oslo Gardermoen Airport, and Stockholm Arlanda Airport to enable me to substitute a flight without losing too much time, comfort, and reliability in my journey.

As a passenger, the current issue is that air transport between CPH, OSL, and ARN is both frequent and fast, but rail transport is too slow and less integrated into the air transport journey. The air-rail service should thus enable passengers to make a rail journey between CPH, OSL, and ARN within a total of no more than 5.5 hours, as beyond that point rail transport is less attractive as a substitute for air transport for short-haul flights. Additionally, rail transport between the airports should be direct in order to increase its attractiveness and reduce uncertainty for passengers. This goes especially for passengers carrying luggage or for long-haul transfer passengers.

In addition, the passenger also expects the service to work as part of one cohesive journey, not as separate transport legs. This implies that the rail leg of the journey should be well synchronized in terms of time

with the flight, should have enough frequency to replace the existing short-haul flights, and should also have enough capacity to replace the existing short-haul flights. In the case of the connecting passenger, the time should still be short but also realistic, such as the time it takes to complete the airport procedures such as check-in, customs, and security. Lastly, as the rail service should also be attractive to the non-transfer passenger, the rail should not only link the airports but also the city centers of Copenhagen, Oslo, and Stockholm, thus enabling airport-city as well as airport-airport travel in one cohesive journey.

Airport operator perspective

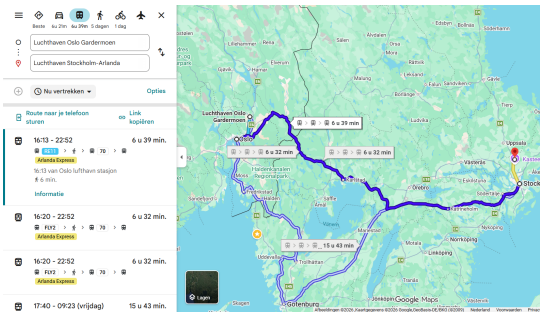
As an airport operator in the Nordic Triangle, I wish the airport to be multimodal hub where rail can replace short-haul flights in a cohesive manner in order to maintain connectivity, enhance sustainability, preserve passenger flows, and also remain commercially viable for future mobility needs.

From an airport operator's point of view, it is not only an environmental but also a strategic and economic challenge. This is because airports such as CPH, OSL, and ARN already heavily rely on good connectivity and transfer traffic, as well as commercial revenue generated by these passenger movements. Therefore, replacing these short-haul flights by rail should not reduce the hub value of an airport but rather increase it by incorporating the train into the existing airport processes. This can be achieved by having the trains stop at the airport and by having a sufficient frequency and capacity to enable passengers to still connect with flights operating on long-haul routes and hubs. The airport also requires the service in order to support its own interests. For instance, a successful design should reduce the burden of aviation on short-haul transport and contribute to reduced emissions. However, it should also ensure that passengers continue to move through the environment and use its facilities. Therefore, it is essential for the service to include integrated ticketing systems and passenger information systems in order to ensure that the airport is not bypassed.

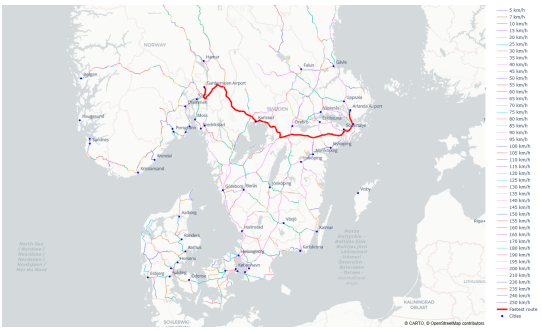
3.4. System KPI's

3.4.1. Journey Time

In the current situation the journey time can be determined by online tools like Google Maps. For example, if one could find the current train journey time between two stations via Google Maps. In the case of designing new rail service, a self-made python tool is developed to measure journey distance and ride time. A more detailed description and explenation of this tool is given in section **??**. A comparison of those KPI measurement tools for journey time are shown in figure 3.2.



(a) Current train route Oslo Gardermoen Airport - Stockholm Arlanda with a ride time of 6 hours and 39 minutes (Google Maps, 2026)



(b) The same route with the custom made rail planner tour with a ride time of 4 hours and 6 minutes. Note that in this case, the ride time does not any intermediate stops

Figure 3.2: Journey time KPI measurements between Google Maps and custom made tool between the route Stockholm Arlanda Airport and Oslo Gardermoen Airport

3.4.2. Transfer Time

The transfer time can measured as the time of arriving / departing train relative to arriving and departing aircraft. This time has a sufficient lower and upper bound. The goal is not necessarily to minimize of maximize this transfer time value, but to find a sufficient transfer time. The transfer time is strongly connected to the way the synchronization rate is calculated.

3.4.3. Number of Transfers

In the current situation the number of transfers can be determined by online tools as Google Maps (as shown in figure 3.2. In figure 3.2 it is shown that the route via rail from Oslo Gardermoen to Stockholm Arlanda has two transfers; from Oslo Gardermoen Airport to Olso Central Station and at Stockholm Arlanda Airport to Stockholm Central Station. For the upgraded design, the number of transfers are kept to a minimum. A transfer is only added in the design if their is a relevant underlying reason.

3.4.4. Synchronization Rate

The synchronization rate is the amount of planes that depart or arrive within a specific time window that overlap with an arriving or departing train. A schematic of such an synchronized system is given in figure 3.3. A benchmark timetable with synchronization can be made using a modified version of the air-rail synchronization optimization model as given by Ke et al., 2020.

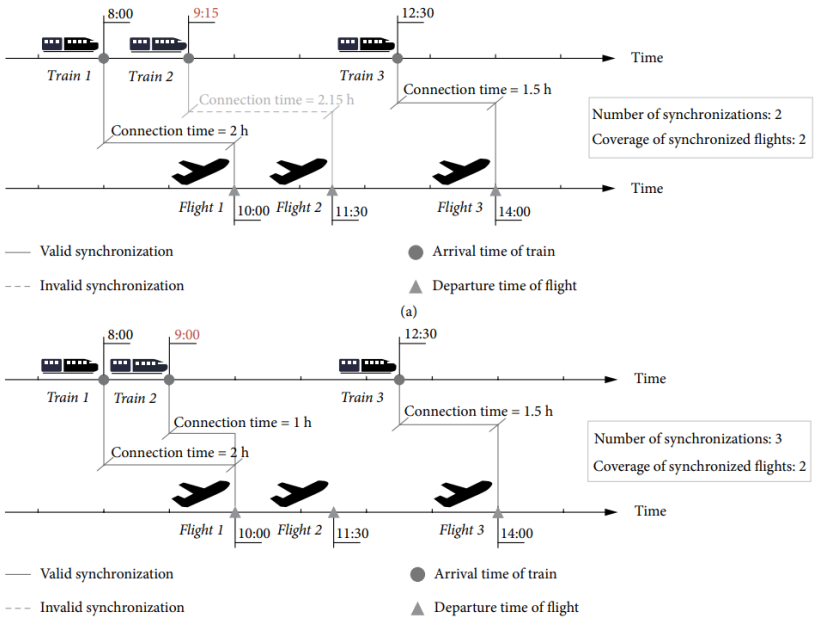


Figure 3.3: Synchronization rates of planes and trains (Ke et al., 2020)

References

Airports, S. (2026). Trafikstatistik på swedavias flygplatser. Retrieved February 27, 2026, from https://www.swedavia.se/contentassets/40c868cfca874b1e8fe5d20c22cbbc97/swedavia_202512.pdf

AS, A. (2016). Rekordhøy kollektivandel i tilbringertransporten til oslo lufthavn. https://web.archive.org/web/20170823160611/https://www.mynewsdesk.com/no/oslo-lufthavn/news/rekordhoey-kollektivandel-i-tilbringertransporten-til-oslo-lufthavn-160361

AS, A. (2025). Trafikkstatistikk 2025 - årsstatistikk. https://om.avinor.no/om-oss/trafikkstatistikk/

Branković, N., & Kalem, A. (2021). Infrastructure costs and benefits of European high-speed rail. Science Engineering and Technology, 1(2), 47–53. https://doi.org/10.54327/set2021/v1.i2.10

Bruno, F., Zefreh, M. M., Fröidh, O., & Cats, O. (2025). Replacing short-haul flights with train travel: Exploring impacts, capacity requirements and policy implications. Transport Policy.

Buire, C., Marzuoli, A., Delahaye, D., & Mongeau, M. (2023). Air–rail timetable synchronisation: Improving passenger connections in Europe within and across transportation modes. Journal of Air Transport Management, 115, 102526. https://doi.org/10.1016/j.jairtraman.2023.102526

Checkland, P. (1999, September). Systems thinking, systems practice. Wiley.

Chiambaretto, P., Baudelaire, C., & Lavril, T. (2013). Measuring the willingness-to-pay of air-rail intermodal passengers. Journal of Air Transport Management, 26, 53. https://doi.org/10.1016/j.jairtraman.2012.10.003

Commission, E., for Mobility, D.-G., & Transport. (2025). Multimodal digital mobility service – report. Publications Office of the European Union. https://doi.org/doi/10.2832/0079219

Consultancy, N. I. A. (2026). About us. Retrieved February 19, 2026, from https://www.naco.nl/en/about-us

Copenhagen Airports A/S. (2025). Passenger and profit growth at copenhagen airport. https://www.cph.dk/en/aboutph/press/news/2025/03/passenger%5C%20and%5C%20profit%5C%20growth%5C%20at%5C%20copenhagen%5C%20airport

Eurostat. (2025). Detailed air passenger routes by country (avia-par). Retrieved February 13, 2026, from https://ec.europa.eu/eurostat/databrowser/explore/all/transp?sort=category&lang=en&subtheme=avia.avia_pa.avia_par&display=list

for the Environment, D. C., & Infrastructure. (2020). Changing tracks [Towards better international passenger transport by train].

Givoni, M., & Banister, D. (2006). Airline and railway integration. Transport Policy, 13(5), 386–397. https://doi.org/10.1016/j.tranpol.2006.02.001

Google Maps. (2026). Google Maps: Oslo Gardermoen to Stockholm Arlanda. https://www.google.com/maps/dir/Luchthaven+Oslo+Gardermoen,+Edvard+Munchs+veg,+2061+Gardermoen,+Noorwegen/Luchthaven+Stockholm-Arlanda,+190+45+Stockholm-Arlanda,+Zweden/@59.6724503,13.1631214,7z/data=!4m16!4m15!1m5!1m1!1s0x4641838b248a1cbf:0xcca0add4a7eb5b82!2m2!1d11.0963701!2d60.1978446!1m5!1m1!1s0x465f9d63c32d83fd:0x52f895a27e4e5c33!2m2!1d17.9342942!2d59.6493928!2m1!5e2!3e3?entry=ttu&g_ep=EgoyMDI2MDMwMi4wIKXMDSoASAFQAw%3D%3D

IATA. (2026). Airline and airport code search. Retrieved February 26, 2026, from https://www.iata.org/en/publications/directories/code-search/

ICAO. (2026). Icao library. Retrieved February 26, 2026, from https://elibrary.icao.int/home?

Janic, M. (2010). Assessing some social and environmental effects of transforming an airport into a real multimodal transport node. Transportation Research Part D Transport and Environment, 16(2), 137–149. https://doi.org/10.1016/j.trd.2010.10.002

Kang, L., Zhu, X., Sun, H., Wu, J., Gao, Z., & Hu, B. (2018). Last train timetabling optimization and bus bridging service management in urban railway transit networks. Omega, 84, 31–44. https://doi.org/10.1016/j.omega.2018.04.003

Ke, Y., Nie, L., Liebchen, C., Yuan, W., & Wu, X. (2020). Improving Synchronization in an Air and High-Speed Rail Integration Service via Adjusting a Rail Timetable: A Real-World Case Study in China. Journal of Advanced Transportation, 2020, 1–13. https://doi.org/10.1155/2020/5081315

Markus Hjortsman, K. S. (2024). Bra lämnar bromma flygplats för arlanda – ska flyga för sas. SVT Nyheter.

OpenRailwayMap. (n.d.). https://www.openrailwaymap.org/

Orban, A. (2025, August). Copenhagen airport reports record passenger numbers, higher profit, and expanding route network in first h ... Retrieved February 27, 2026, from https://www.aviation24.be/airports/copenhagen/copenhagen-airport-reports-record-passenger-numbers-higher-profit-and-expanding-route-network-in-first-half-of-2025/

Parliament, T. E., & Council. (2016). Directive (eu) 2016/797 of the european parliament and of the council on the interoperability of the rail system within the european union. EUR-Lex.

Rattanakijsuntorn, W., Suwannarat, B., Samittivate, N., & Nithikittiwat, C. (2024). Success Factors and Partnership Evaluation of Air–Rail Integration Development: A Case of a High-Speed Rail Project Linking Three Airports in Thailand. Infrastructures, 9(7), 115. https://doi.org/10.3390/infrastructures9070115

Román, C., & Martín, J. C. (2014). Integration of HSR and air transport: Understanding passengers' preferences. Transportation Research Part E Logistics and Transportation Review, 71, 129–141. https://doi.org/10.1016/j.tre.2014.09.001

Royal HaskoningDHV. (2025, August). Onderzoek internationaal spoorvervoer corridors.

SAS. (2024). SAS Annual and Sustainability Report – Fiscal Year 2024. https://www.sasgroup.net/sustainability/sustainability-reports/sas-annual-and-sustainability-report-fiscal-year-2024/

Toet, A., Van Kuijk, J., Boersma, K., & Santema, S. (2025). Tensions in air and rail integration based on a European longitudinal case study with stakeholders. npj Sustainable Mobility and Transport, 2(1). https://doi.org/10.1038/s44333-025-00062-4

Tove Andersson, D. L. (2013). Station design on high speed railway in scandinavia [Master's thesis, Chalmers University of Technology Göteborg, Sweden].

van Dam, J. (2024). Towards air and rail integration - assessing the preferences of dutch travellers for multi-legged journeys. TU Delft.

Wen, F., Chen, Y., Bai, Y., Zhu, Q., & Li, N. (2024). Urban rail train timetabling for the end-of-service period with passenger accessibility and operation cost: An advanced benders decomposition algorithm. Transportation Research Part B Methodological, 190, 103094. https://doi.org/10.1016/j.trb.2024.103094

Xia, W., & Zhang, A. (2017). Air and high-speed rail transport integration on profits and welfare: Effects of air-rail connecting time. Journal of Air Transport Management, 65, 181–190. https://doi.org/10.1016/j.jairtraman.2017.06.008